1. The smallest size is int8 with 5.76kb. The model that performed the best all the models, with an accuracy of .98.
2. My proposed strategy to reduce the size of the model is to prune with an int8 quantization.
3. Results: The combination of pruning with int8 quantization did reduce the model size significantly compared to all the other models, however, the int8 model still outperformed my proposed method by a little bit. Thankfully the accuracy did not go down but combining both int8 quantization and pruning did not achieve the result I wanted in the end.

Problem 2:

1. The learning blocks available under the free-tier edge impulse account are : (in the format of Learning blocks : example model)
    a. Classification(keras): Neural Network
    b. Regression (keras) : Keras Regression model
    c. Anomaly Detection(k-means): K-means anomaly detector
    d. Anomaly Detection(GMM): Gaussian Mixture Model
    e. Visual anomaly detection(FOMO-AD): FOMO-Ad visual anomaly detector
    f. Image classification(using transfer learning): MobileNetV2 image classifier
    g. keyword spotting (using transfer learning): CNN keyword spotting model
    h. object detection(using mobilenetv2 ssd FPN): MobileNetV2 SSD Detector
    i. Object detection(using FOMO), classical ML).: FOMO Detector
2. Types of layers:
    a. Input layer: Received the input features from the processing block
    b. Dense Layer: Connects every neuron to all the neurons in the preceding layer
    c. Convolution layers: takes spatial information and extracts the patterns
    d. Reshape: Changes dimensions to match required shape between layers
    e. Flatten: flattens or converts outputs into one-dimensional vector
    f. Dropout: Randomly cuts a fraction of network connections during training
    g. Pooling layer: reduces the feature sizes and complexity while retaining important information
    h. Output layer: produces the final prediction
3. Model Compression options :
    a. Quantized (int8)
    b. Unoptimized(float32)
    c. EON Compiler
    d. TensorFlow Lit
4. Input modalities supported :
    a. Image data:
        i. Uses flux-pro image generation to generate synthetic images
        ii. DALL-E synthetic image generator
    b. Audio/voice data:

    i. Whisper Synthetic voice generator

c. It is important for machine learning because it can handle real-world data limitations, reduces overfitting and can improve model generalization.